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BRIEF REPORT

The Concurrent Validity of Session-Rating of Perceived Exertion Workload Obtained Face-to-Face Versus Via an Online Application: A Team Case Study

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Purpose: To compare the concurrent validity of session-rating of perceived exertion (sRPE) workload determined face-to-face and via an online application in basketball players. ***Methods:*** Sixteen semiprofessional, male basketball players (21.8 [4.3] y, 191.2 [9.2] cm, 85.0 [15.7] kg) were monitored during all training sessions across the 2018 (8 players) and 2019 (11 players) seasons in a state-level Australian league. Workload was reported as accumulated PlayerLoad (PL), summated-heart-rate-zones (SHRZ) workload, and sRPE. During the 2018 season, rating of perceived exertion (RPE) was determined following each session via individualized face-to-face reporting. During the 2019 season, RPE was obtained following each session via a phone-based, online application. Repeated-measures correlations with 95% confidence intervals were used to determine the relationships between sRPE collected using each method and other workload measures (PL and SHRZ) as indicators of concurrent validity. ***Results:*** Although all correlations were significant ($P < .05$), sRPE obtained using face-to-face reporting demonstrated stronger relationships with PL ($r = .69$ [.07], large) and SHRZ ($r = .74$ [.06], very large) compared with the online application ($r = .29$ [.25], small [PL] and $r = .34$ [.22], moderate [SHRZ]). ***Conclusions:*** Concurrent validity of sRPE workload was stronger when players reported RPE in an individualized, face-to-face manner compared with using a phone-based online application. Given the weaker relationships with other workload measures, basketball practitioners should be cautious when using player training workloads predicated on RPE obtained via online applications.

Keywords: training load, team sport, court-based sport, RPE, microsensors

In basketball, monitoring player workloads is important in managing training prescription to promote positive performance-related adaptations in players.[1] Despite many approaches being available for workload monitoring in basketball, the session rating of perceived exertion (sRPE) is one of the most frequently utilized measures in practice.[2] In this regard, sRPE offers a number of practical advantages to basketball coaching staff compared with other approaches given it is simple, cost-effective, and noninvasive to players. Furthermore, the subjective-based sRPE has been validated against objective measures of exercise intensity.[3]

The accuracy of sRPE is, however, contingent on a number of factors such as use of validated scales, collecting ratings of perceived exertion (RPE) from each player in the absence of peers, and appropriate familiarization processes.[4] Further to these points, RPE has been traditionally collected via face-to-face reporting whereby players provide individualized ratings to a basketball practitioner following training or games. However, given the increased use of smart devices among athletes[5] and the growing availability of suitable applications to collect monitoring data, measurement of RPE via phone-based, online applications is increasing. Furthermore, use of applications may be practically beneficial in allowing multiple subjective metrics (eg, wellness scores, injury symptoms) to be collected and stored using a central online platform. Despite these practical advantages, it is not clear whether collection of RPE via online applications to determine sRPE workload is equally valid as conventional face-to-face reporting.

Evidence regarding the concurrent validity of sRPE determined using an online application compared with other workload measures is limited and only provided for water polo[5] and endurance athletes.[6] Nevertheless, initial findings suggest acceptable validity when compared with Banister's training impulse ($r = .51–.79$)[5] and summated-heart-rate-zones (SHRZ) workload ($r = .54–.83$[5] and $r = .80$[6]). While promising, further work is necessary to gather a more definitive consensus on the topic and determine whether similar results exist in basketball where training demands may differ to other sports. Therefore, the purpose of this study was to quantify and compare the concurrent validity of sRPE collected face-to-face and using an online application in a basketball team.

Methods

Subjects

A total of 16 semiprofessional, male basketball players volunteered to participate in the study, including 8 (age: 23.1 [3.8] y; stature: 191.4 [8.1] cm; body mass: 87.5 [15.5] kg) monitored during the 2018 season, and 11 (21.4 [4.5] y; 191.8 [10.3] cm; 86.4 [20.6] kg) monitored during the 2019 season. All players were registered to the same team in the Queensland Basketball League, which is a second-tier, Australian basketball competition. Players provided voluntary written informed consent along with parental consent (if aged <18 y) prior to participation. All study protocols were approved by the Central Queensland University Human Research Ethics Committee.

Design

An observational study design was implemented whereby players were monitored during training sessions in 2 separate seasons.

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Players participated in 1 to 3 training sessions per week with 163 and 57 samples obtained for the 2018 and 2019 seasons, respectively.

Methodology

Prior to each training session, players were fitted with microsensors (OptimEye S5; Catapult Innovations, Melbourne, Australia) held at the upper torso between the scapulae. Heart rate (HR) data were measured via chest-worn straps (Polar T31; Polar Electro Oy, Kempele, Finland) during the 2018 season and via shirts (Polar Team Pro Shirt; Polar Electro Oy) during the 2019 season. Agreement in HR between these methods is acceptable ($r = .99$).[7]

Players were familiarized with the RPE scale prior to study commencement. Following each session, players reported their RPE using Borg 0 to 10 category ratio scale. During the 2018 season, players reported their RPE individually to the same researcher after seeing a printed scale. Alternatively, during the 2019 season, players reported RPE using an online application (Team App, www.teamapp.com) on their personal smartphone. In the application, RPE was collected using a customizable questionnaire (google forms, www.docs.google.com/forms) to replicate the printed scale content and formatting, therefore, limiting potential bias between seasons. Players reported their RPE within 30 (face-to-face) and 60 (online) min of completing each session.

External workload was derived from the accelerometer component of the microsensor and reported as accumulated PlayerLoad™ (PL). The PL is calculated as the square root of the sum of the squared rate of change in acceleration across the transverse (x), coronal (y), and sagittal (z) planes multiplied by a scaling factor of 0.01. The reliability of PL (coefficient of variation = 0.9%–1.9%) in team sports has been supported.[8]

Following training, HR data were exported in 1-s epochs into a specifically designed spreadsheet for analysis (Microsoft Excel version 15; Microsoft Corp, Redmond, WA). Internal workload was reported as SHRZ workload[9] whereby HR responses were placed into predefined zones spanning 10% between 50% and 100% of maximum HR (HR_{max}) taken as the highest HR from any training session or game.[10] Time (minutes) in each zone was then multiplied by corresponding weightings from 1 (ie, 50%–60% HR_{max}) to 5 (ie, 90%–100% HR_{max}) with workload calculated as the sum of weightings from each zone. Finally, sRPE was determined as individualized RPE multiplied by session duration (minutes).[11]

Statistical Analysis

Each workload measure is reported as mean (SD). To determine relationships between sRPE and PL as well as SHRZ, repeated-measures correlations (r) with 95% confidence intervals were calculated, with data from each season reported separately to compare RPE methods. Repeated-measures correlations were appropriate given the nonindependence of samples (ie, multiple observations per player). Correlation magnitudes were interpreted as trivial, <0.10; small, 0.10 to 0.29; moderate, 0.30 to 0.49; large, 0.50 to 0.69; very large, 0.70 to 0.89; nearly perfect, 0.90 to 0.99; and perfect, 1.00. All analyses were conducted using the "rmcorr" package in R Studio (version 3.5.3; R Foundation for Statistical Computing, Vienna, Austria). Significance was accepted where $P < .05$.

Results

Mean (SD) workloads across each season are presented in Table 1. Correlation plots between sRPE using face-to-face and online application reporting and PL as well as SHRZ are presented in Figure 1. For face-to-face reporting, significant ($P < .001$) correlations were found between sRPE and PL (large) and SHRZ (very large). When using the online application, correlations between sRPE and PL (small) and SHRZ (moderate) were also significant ($P < .05$), but weaker than face-to-face reporting.

Table 1 Mean (SD) Workloads Encountered During Training Across Each Season in a Basketball Team

Workload measure	2018 season	2019 season
PL, AU	454 (123)	457 (110)
SHRZ workload, AU	212 (71)	230 (67)
sRPE,[a] AU	387 (175)	423 (140)

Abbreviations: AU, arbitrary units; PL, PlayerLoad; SHRZ, summated heart-rate zones; sRPE, session rating of perceived exertion.
[a]Rating of perceived exertion was collected via face-to-face reporting in 2018 and using an online application in 2019.

Discussion

Establishing concurrent validity between objective and subjective workload measures is necessary in assessing whether similar constructs are being measured, which can assist in making informed choices on monitoring choices. Our data revealed sRPE was more strongly correlated with PL and SHRZ when RPE was reported face-to-face compared with using an online application. These findings contrast previous research[5,6] documenting large–very large correlations between sRPE collected via an online application and HR-based measures using Banister's training impulse and SHRZ workload. It is difficult, however, to directly compare our results to past work,[5,6] given differences in the samples investigated (water polo and endurance athletes vs basketball players). For example, basketball is an intermittent, court-based team sport stressing anaerobic and aerobic pathways. In this regard, physical demands differ from sports such as water polo, which is not weight-bearing and endurance activity, which relies primarily on aerobic metabolic energy provision. Consequently, the physical and physiological demands encountered and subsequent perception of training demands is not comparable with aquatic or endurance sports.

The weaker concurrent validity we observed for sRPE collected via an online application may be due to the presence of confounding variables not apparent when reporting RPE face-to-face. For example, collection of RPE in the absence of peer influence is necessary for unbiased reporting.[12] In turn, when players report RPE on personal smartphones, it becomes more difficult to enforce this requirement. In addition, it is possible that players felt more pressure to accurately report their RPE in the presence of the research team during face-to-face reporting.[13] Although the importance of accurate, nonbiased reporting of RPE was explained to players, players may have been more accountable for their data when reporting directly to the research team than when using the online application.[13] Consequently, conventional face-to-face reporting may be more appropriate for collection of RPE data in basketball for determination of sRPE workload.

Although the present study poses novel insight, some limitations were encountered. First, although data were collected in the same team, changes in playing rosters meant only 3 players were monitored across both seasons. Second, compliance was higher where RPE was reported face-to-face compared with online,

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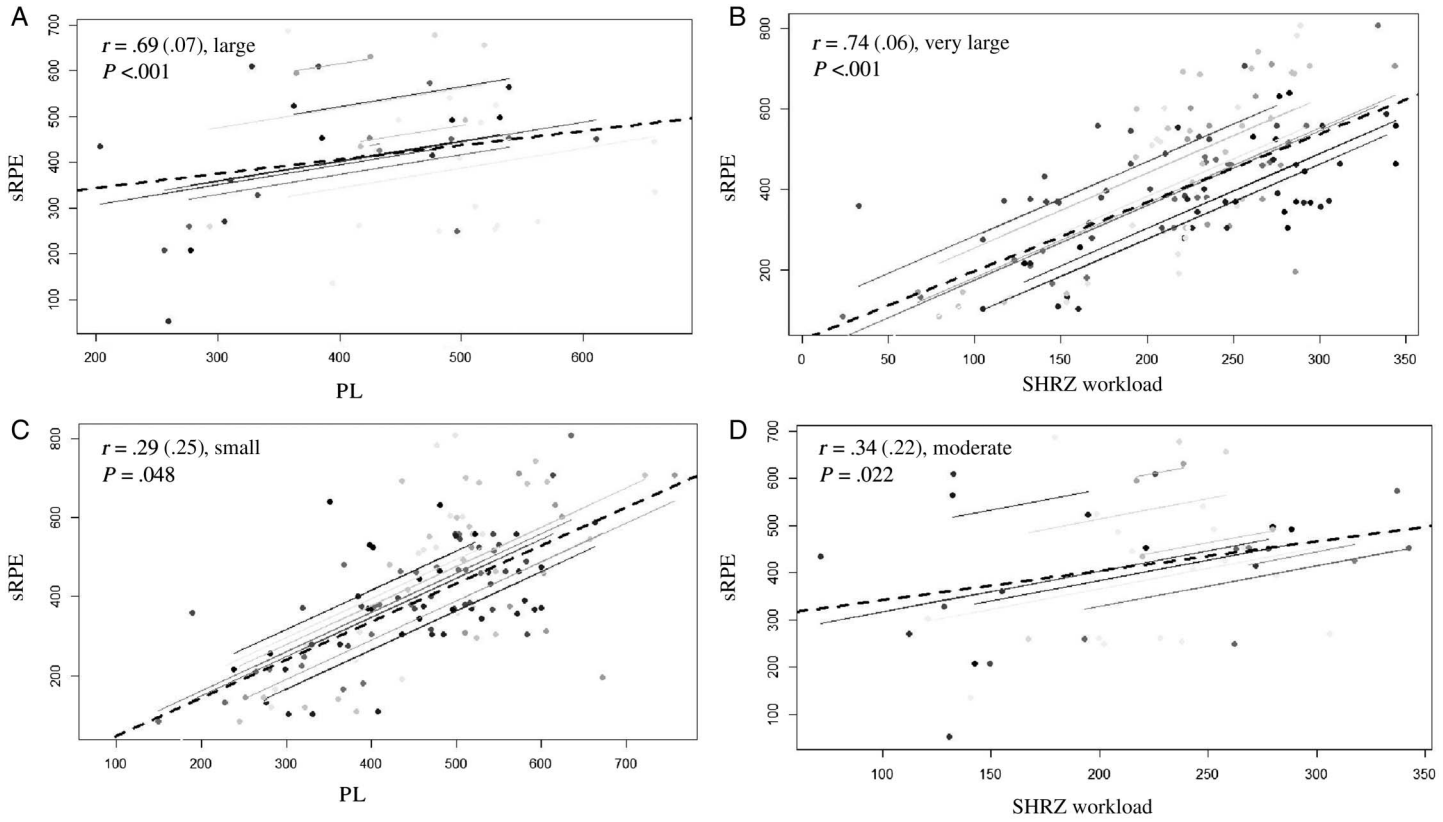

Figure 1 — Repeated-measures correlations (±95% CIs) between sRPE using face-to-face reporting and PL (A), as well as SHRZ workload (B), and using an online application and PL (C), as well as SHRZ workload (D). CI indicates confidence interval; sRPE, session rating of perceived exertion; PL, PlayerLoad; SHRZ, summated heart-rate zones.

subsequently leading to differences in the number of samples obtained in each season. Nevertheless, variations in compliance between methods is an important practical outcome that basketball coaching staff should consider when deciding on the implementation of face-to-face or online reporting of RPE for player monitoring.

Practical Applications

Where possible, the use of face-to-face reporting for determination of sRPE in basketball should be utilized. In addition, strategies to improve the concurrent validity of online RPE in applied settings should be considered given the practical advantages of this method.

Conclusions

Session-rating of perceived exertion determined via an online application possessed small–moderate relationships with PL and SHRZ in basketball players. In contrast, sRPE determined using conventional face-to-face reporting possessed large–very large relationships with PL and SHRZ, demonstrating a stronger concurrent validity than when determined using online reporting.

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